LD11 Final Project Summary

Detailed overview of objectives, datasets, practical work, results, and recommendation.

Field	Value
Student	Ruturaj Mokashi
Topic	Learning Day 11 - Hyperparameter Tuning
Document type	Individual PDF documentation
Date	2026-06-14

Submission note
This is an individual PDF document. The documentation package intentionally does not include a combined/merged PDF.

1. Purpose of the Extended Submission

The main Day 11 notebook covered the required assignment tasks. This extended notebook and documentation package were created because the uploaded source material contained additional topics that were not fully represented by the MVD. The purpose is to close that gap with a stable, local-safe, university-ready submission supplement.

The extended notebook avoids fragile execution. It uses built-in scikit-learn datasets, short commented code cells, clear theory sections, and output-based conclusion cells. This makes the notebook suitable for local execution without internet, package installation, or kernel-killing deep-learning dependencies.

Scope correction
The MVD is the minimum requirement, not the full source checklist. This package documents both the assignment coverage and the additional upl

2. Practical Work Completed

Area	Dataset	Method	Result / purpose
Classification tuning	Breast Cancer Wisconsin	KNN GridSearchCV	Accuracy 0.9790, F1 0.9836, ROC-AUC 0.9923
Linear classification tuning	Breast Cancer Wisconsin	Logistic Regression GridSearchCV	Accuracy 0.9860, F1 0.9889, ROC-AUC 0.9971
Regression tuning	Diabetes	Ridge Regression GridSearchCV	RMSE 53.2042, MAE 41.4595, R2 0.4881
CASH	Breast Cancer Wisconsin	Model family selection	Logistic Regression selected as best family
Advanced optimisation	Breast Cancer Wisconsin	Hyperopt-style Random Forest search	ROC-AUC 0.9933
Advanced optimisation	Breast Cancer Wisconsin	Optuna-style Random Forest search	ROC-AUC 0.9931
Multi-fidelity optimisation	Breast Cancer Wisconsin	Successive Halving Random Forest	ROC-AUC 0.9916

3. Final Result Interpretation

The strongest classification result in the extended notebook is Logistic Regression tuned with GridSearchCV. It achieved ROC-AUC 0.9971 and F1-score 0.9889. This is important because it shows that a simple model can outperform more complex search methods when the dataset is well suited to linear separation.

KNN also performed strongly, but it is distance-sensitive and depends heavily on scaling. Random Forest searches through Hyperopt, Optuna, and Successive Halving were included because the uploaded sources covered advanced optimisation methods. They are important learning topics, even though they did not beat Logistic Regression on this small stable dataset.

Ridge Regression was included to cover regression hyperparameter tuning, which was missing from the first version. Its result is moderate because the diabetes dataset is harder and more linear-regression limited. The point of this section is to show the complete tuning process for a regression model, not to claim production-grade prediction quality.

4. Final Recommendation

For the extended classification workflow, use Logistic Regression GridSearchCV as the best final model because it provides the strongest metrics and the clearest explanation. Use KNN as a strong alternative baseline. Use Hyperopt, Optuna, and Successive Halving as advanced tuning references that demonstrate how different search strategies behave.

Final decision

Best practical model: Logistic Regression GridSearchCV. Best learning value: the full comparison across classical tuning, CASH, Hyperopt, Optun